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Introduction to audio coding

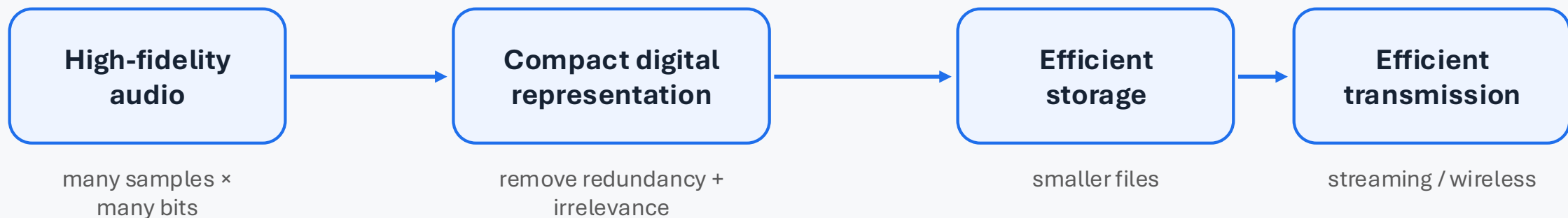
From digital audio to perceptual compression

Based on Spanias, Painter & Atti,
Audio Signal Processing and Coding,
Chapter 1

Core question: how can we reduce bits without changing what listeners perceive?

Why do we need audio coding?

Digital audio is high quality, but raw PCM consumes bandwidth and storage.



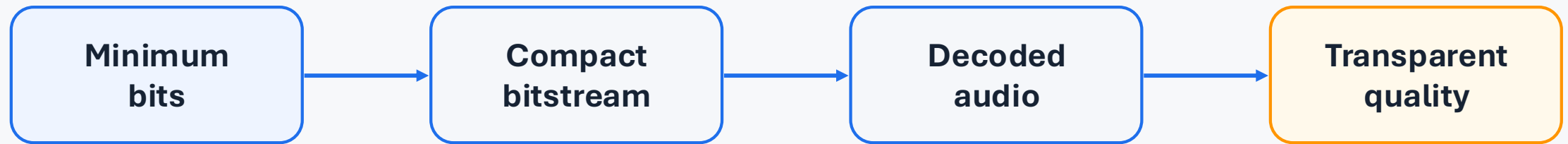
Audio coding is the engineering compromise between rate, quality, delay, and robustness.

Learning goals for this introduction

- 1 Compute raw PCM bit rates from f_s , bit depth, and channels.
- 2 Explain why sampling and quantization are not compression by themselves.
- 3 Distinguish lossless, waveform, perceptual, parametric, and neural codecs.
- 4 Read a generic perceptual encoder block diagram.
- 5 Evaluate codecs using quality, bit rate, complexity, delay, and robustness.

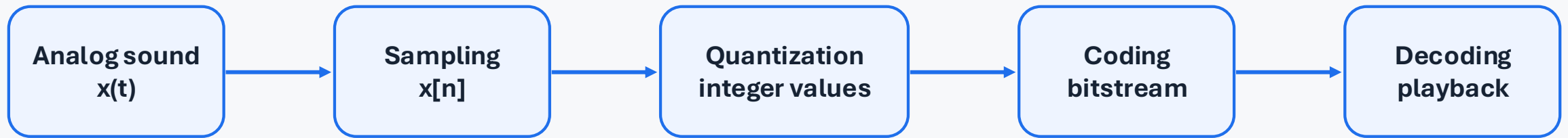
Audio coding in one sentence

Represent audio with the minimum number of bits while keeping the decoded signal perceptually indistinguishable from the original.



“Transparent” means the listener cannot reliably distinguish coded from original audio.

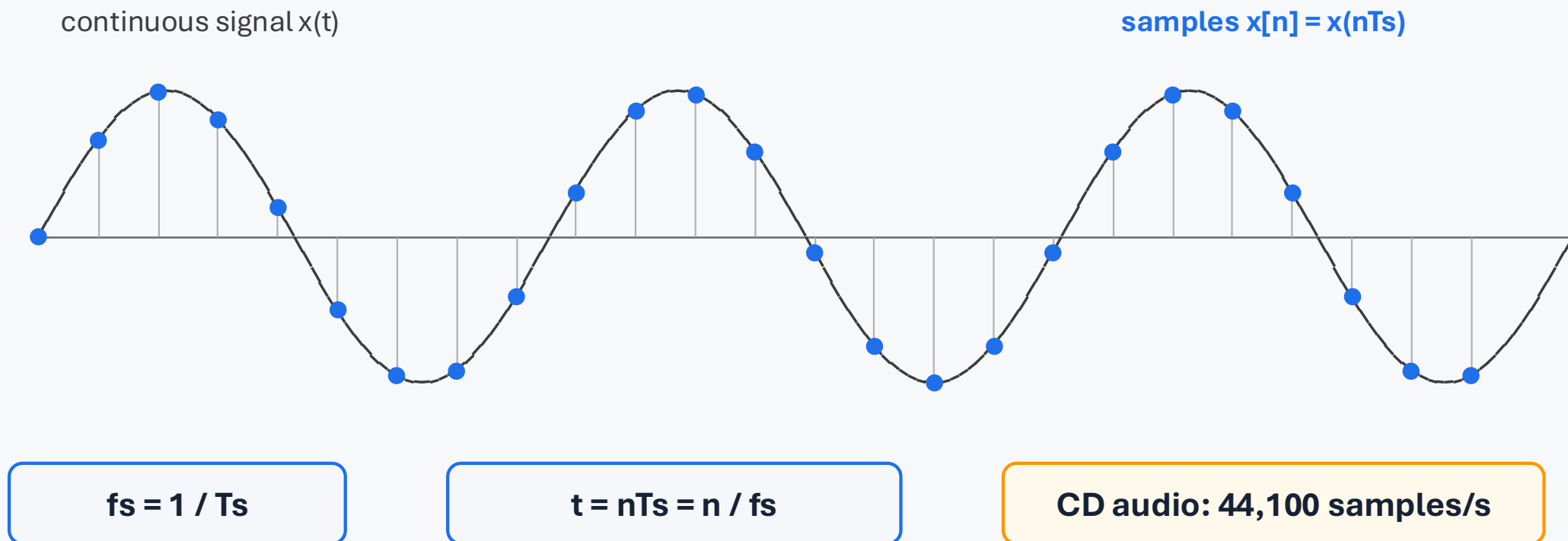
From acoustic pressure to a stream of bits



Sampling discretizes time. Quantization discretizes amplitude. Coding reduces the number of bits needed to represent the signal.

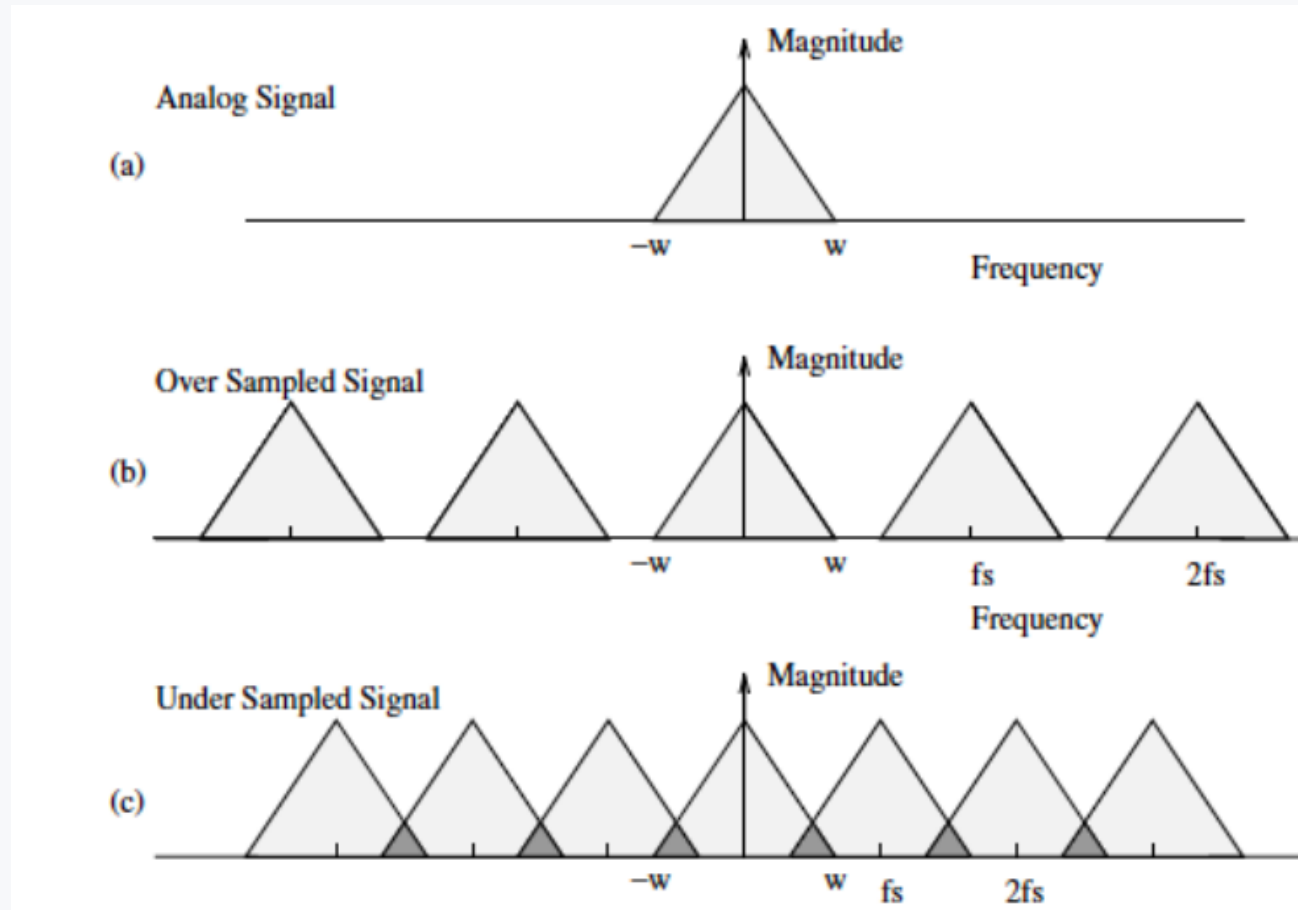
Sampling: time becomes an index

Uniform sampling takes values of a continuous signal at regularly spaced times.



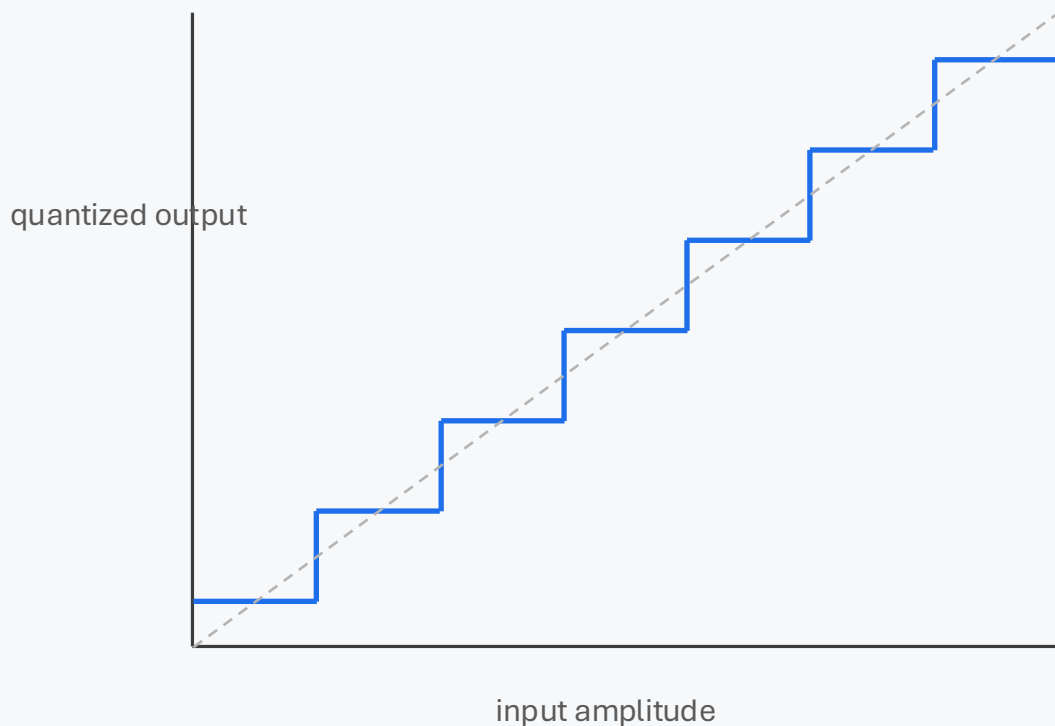
Sampling theorem: avoid spectral overlap

Exact reconstruction is possible when $f_s > 2 f_{\max}$. If spectral replicas overlap, aliasing occurs.



Quantization: amplitude becomes a finite set

A B-bit quantizer maps each sample amplitude to one of 2^B discrete levels.



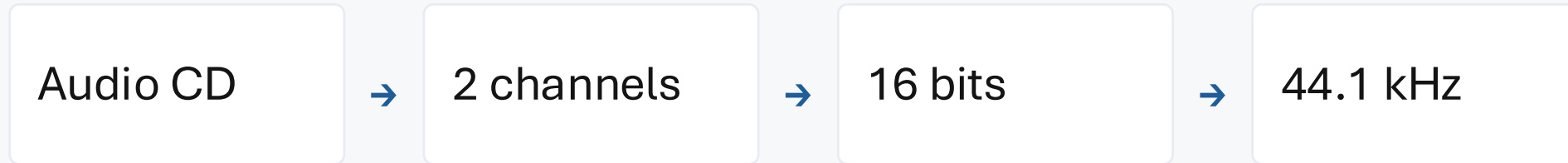
More bits $\rightarrow$ more levels $\rightarrow$ lower quantization noise

But raw PCM rate grows linearly with bit depth B

Perceptual coders place noise where it is least audible

Raw PCM bit rate: the basic calculation

bit rate = channels × bit depth × sampling rate



$$2 \times 16 \times 44,100 = 1,411,200 \text{ bit/s} = 1,411.2 \text{ kbit/s}$$

What can an audio coder remove?

Statistical redundancy
Predictable structure in the signal

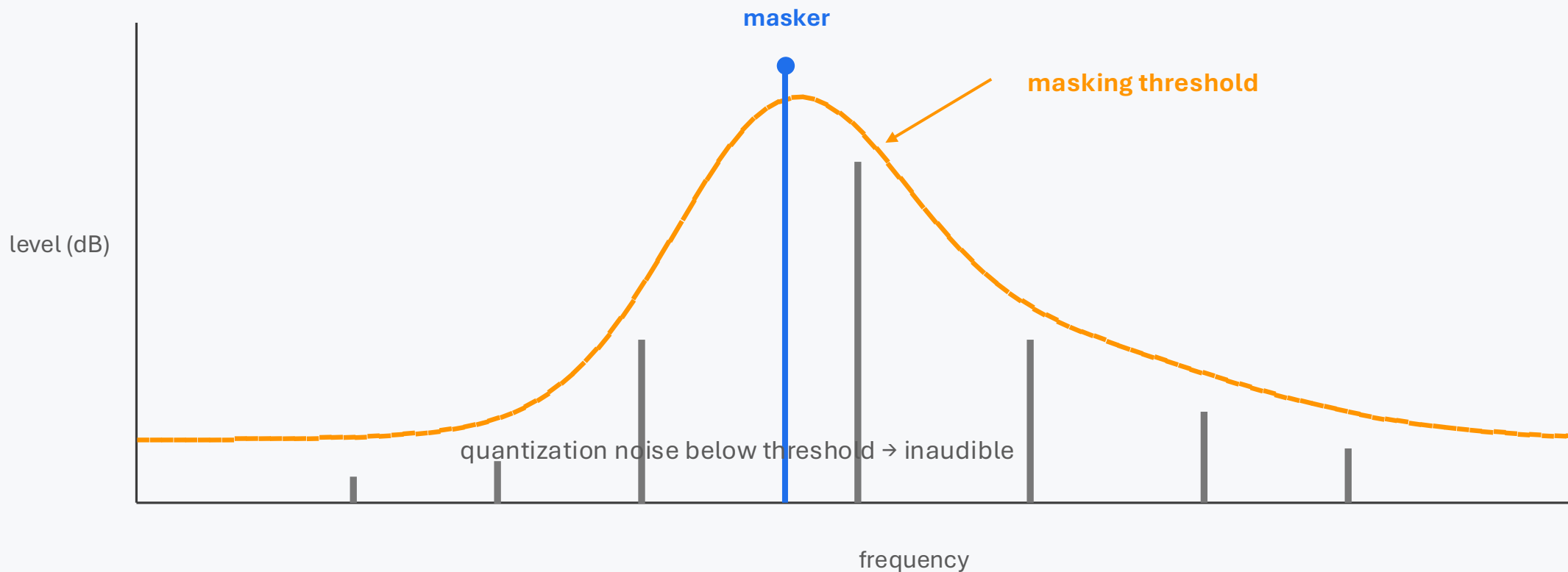
Perceptual irrelevance
Signal components or noise that
listeners do not notice

**Lossless coding removes
redundancy only.
Lossy perceptual coding also
exploits irrelevance.**

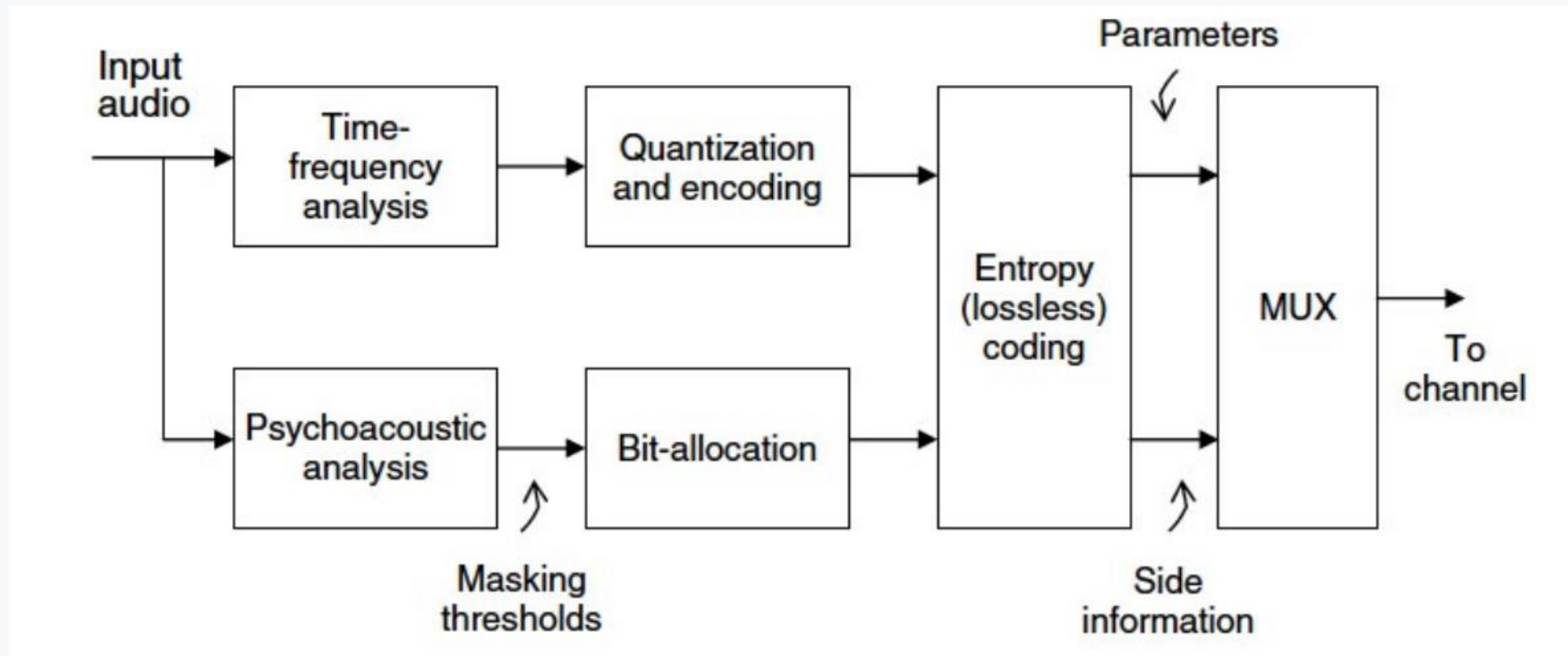
This is why lossy coders can reach
much higher compression ratios, at
the cost of perfect reconstruction.

The perceptual coding principle

Quantization noise is allowed, but it should stay below the masking threshold.

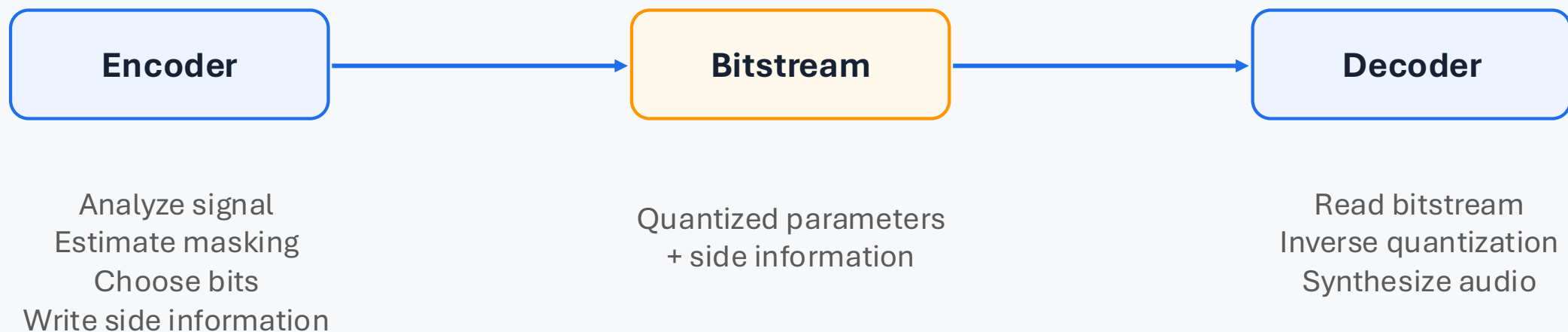


Generic perceptual audio encoder



The psychoacoustic model controls quantization through masking thresholds; entropy coding removes remaining statistical redundancy.

Encoding and decoding are asymmetric



Most complexity is usually in the encoder, where analysis, psychoacoustics and bit allocation are performed.

Audio coder attributes

Quality

Can listeners hear artifacts?

Bit rate

How many kb/s?

Complexity

Can it run in real time?

Delay

Is it usable for interaction?

Robustness

What happens with losses/errors?

Codec choice is application-dependent: music streaming, broadcasting, storage, games, hearing devices, and real-time communication optimize different tradeoffs.

Main families of audio coders

Lossless

Exact reconstruction
Example: FLAC

Predictive

Exploit short/long-term correlation
Often speech-oriented

Transform/subband

Time-frequency coefficients
Used in many MPEG coders

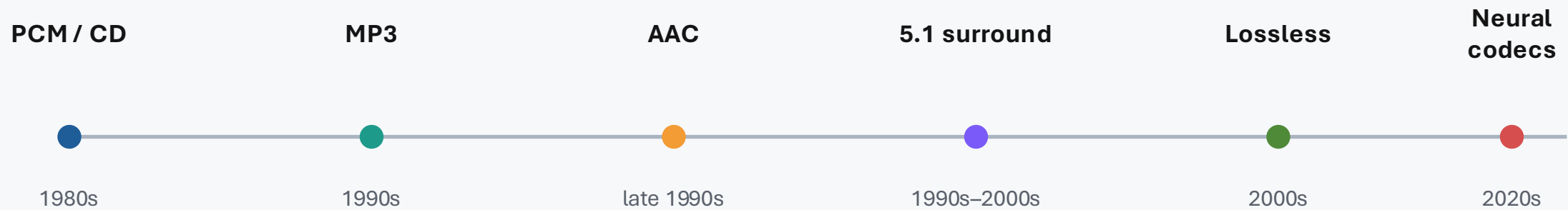
Sinusoidal/parametric

Represent sound by model parameters

Neural

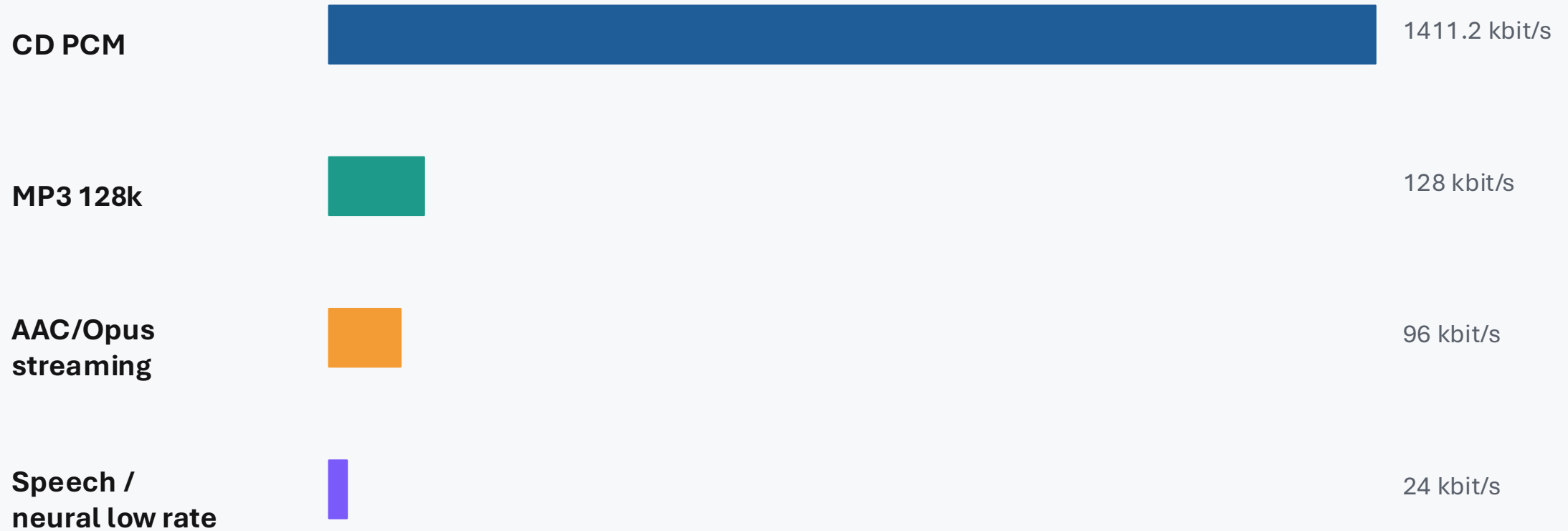
Learned analysis/synthesis and quantization

Standards and formats: a historical arc



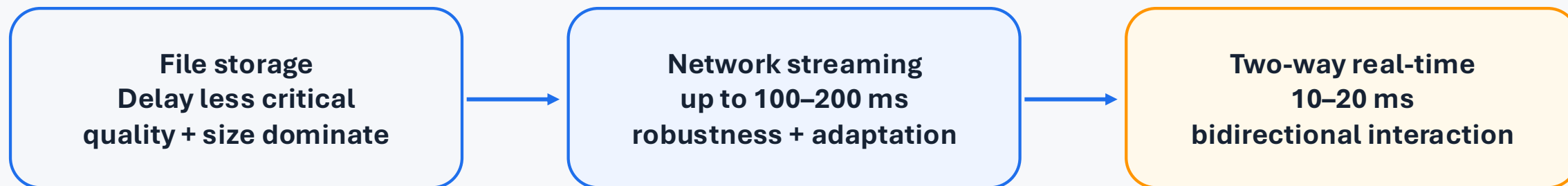
Early perceptual standards grew around multimedia storage and transmission; later systems added multichannel, lossless, low-delay, and learned coding goals.

Bit-rate ladder: compression as a scale



Compression ratio vs CD PCM: MP3 128 kbit/s is about 11:1

Delay matters: choose the codec for the use case



Lower delay usually means shorter frames and less look-ahead, which can reduce coding efficiency.

Suggested class demos

Using the examples in <https://github.com/xserra/audio-coding-materials/>

1

[sound-display.ipynb](#)

Display a waveform and listen to the relation between samples and sound.

2

[downsample-chirp.ipynb](#)

Hear and visualize aliasing when a chirp is downsampled.

3

[resample.ipynb](#)

Compare resampling choices and their effect on spectral content.

4

[mp3-coding.ipynb](#)

Encode audio as MP3 and compare bit rate, file size and artifacts.

Takeaways

Audio coding is about reducing bit rate while preserving perceived quality.

Raw PCM bit rate is simple: $\text{channels} \times \text{bit depth} \times \text{sampling rate}$.

Perceptual coders combine time-frequency analysis, masking models, quantization, bit allocation, and entropy coding.

No codec is universally best: quality, bit rate, complexity, delay, and robustness define the design space.

Reference: Spanias, Painter & Atti (2007), Audio Signal Processing and Coding, Chapter 1.